



CVL2001: Climate Change and Adaptation

Lecture 2: How Do We Know the Climate Is Changing?

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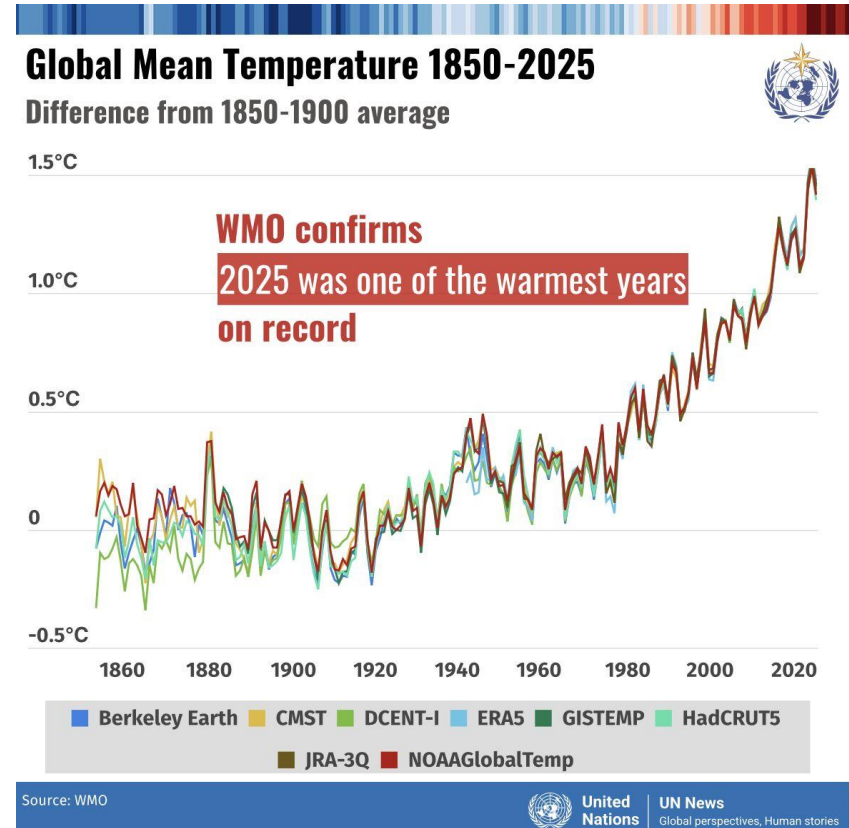
Two Numbers, One Climate Story?

GLOBAL (WMO)

- 2025 was among the three warmest years on record.
- $+1.44^{\circ}\text{C} \pm 0.13^{\circ}\text{C}$ relative to 1850–1900

INDIA (IMD)

- 2025 was India's eighth-warmest year since 1901.
- $+0.28^{\circ}\text{C}$ relative to 1991–2020
- Are these reports telling different stories?



Are These Reports Contradictory?

1. Why are the two temperature anomalies different?
2. Why does 2025 have a different global and Indian ranking?
3. What information should we check before comparing the two reports?

Why the Reports Are Not Contradictory

1. Different geographical coverage

- WMO reports the global average.
- IMD reports the average over Indian land.

2. Different reference periods

- WMO compares 2025 with 1850–1900.
- IMD compares 2025 with 1991–2020.

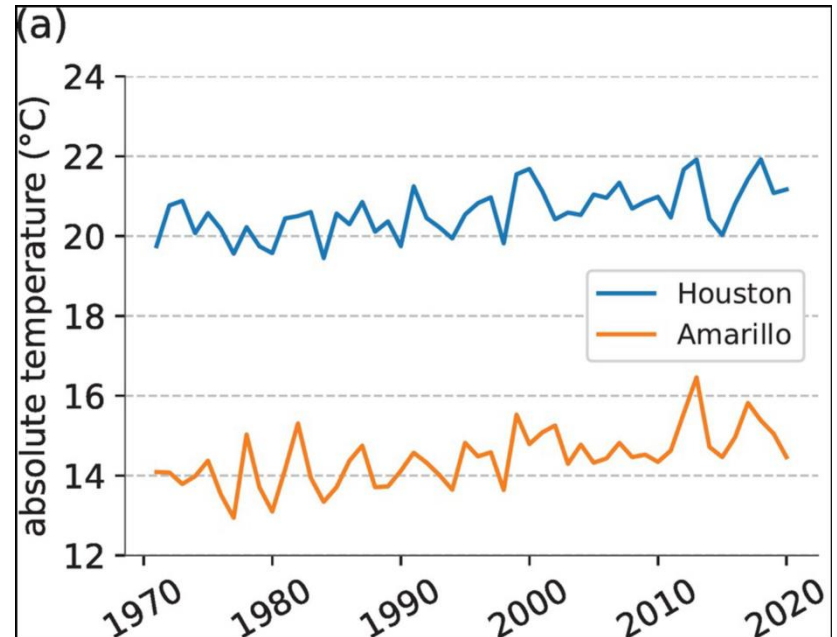
3. Different temperature rankings

- Regional temperatures do not change identically every year.
- Rankings depend on the dataset, geographical coverage and period of record.

Key takeaway: Before comparing the reports, we must understand how their temperature anomalies were calculated.

Why Not Compare Absolute Temperatures?

- Houston is naturally warmer than Amarillo.
- Their annual mean temperatures differ by about 6°C.
- Local elevation, latitude and geography affect absolute temperature.

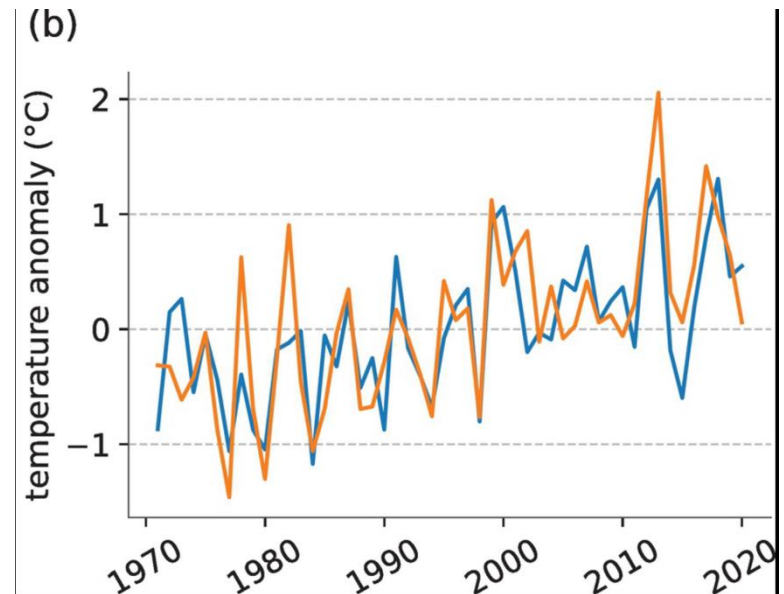


Source: Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.1

Relative Temperatures?

- The graph shows annual temperature anomalies relative to each location's 1980–2010 average.
- Houston and Amarillo have different normal temperatures.
- Warm and cool years often occur in both places together.
- Removing the difference in their normal climates makes the shared pattern easier to see.

Key takeaway: Temperature anomalies allow places with different climates to be compared.



Source: Dessler, A. E. (2021), *Introduction to Modern Climate Change*, 3rd ed., Chapter 2, Figure 2.1

A Temperature Anomaly Is a Difference

Temperature
anomaly

=

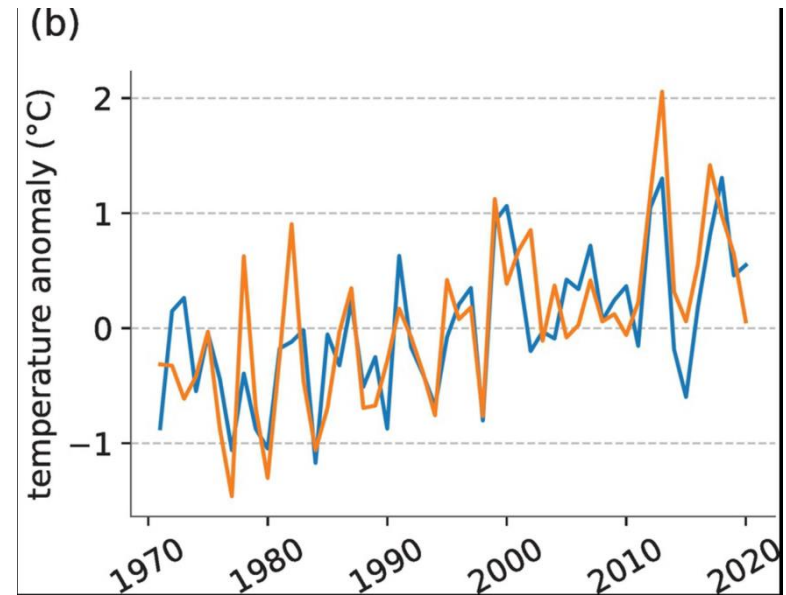
Observed
temperature

−

Reference-period
average

- **Positive:** warmer than the reference average
- **Negative:** cooler than the reference average
- **Zero:** equal to the reference average

It measures the departure from normal—not the actual temperature.



Source: Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.1

Example: Calculating a Temperature Anomaly

Q) A station has:

- Reference-period average: 20°C
- Observed annual average: 21.5°C
- Calculate the temperature anomaly.

Answer: ?

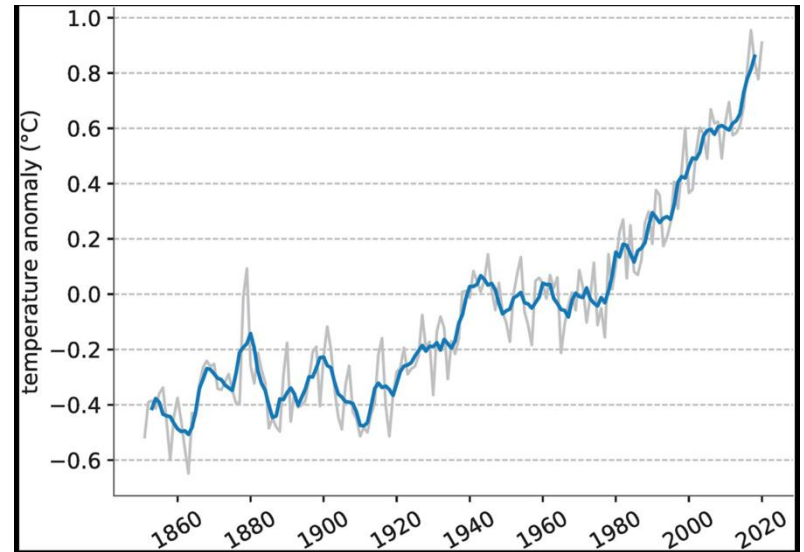
Interpretation: The year was 1.5°C warmer than the station's reference-period average. This does not mean the actual temperature was 1.5°C .

The Reference Period Matters

The same observed temperature can produce different anomalies depending on the reference period.

- **Observed temperature: 21.0°C**
- Reference average: 20.0°C → **Anomaly: +1.0°C**
- Reference average: 20.5°C → **Anomaly: +0.5°C**

Always ask: relative to which reference period?



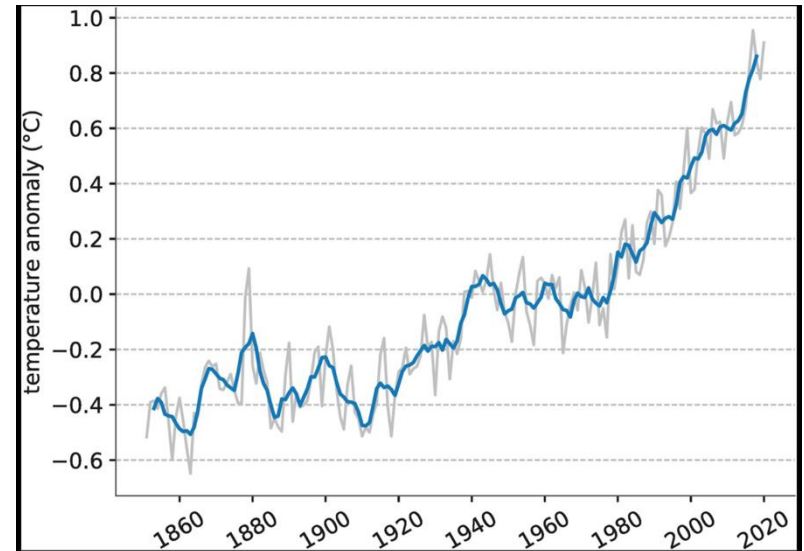
Source: Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.1. Gray shows annual anomalies, white blue shows the smoothed warming trend, relative to the 1951–1980 average.

From Local Variability to a Global Signal

- A single station is strongly affected by local weather.
- When anomalies from many well-distributed stations are averaged:
 - Local warm and cool fluctuations partly cancel.
 - The common large-scale temperature signal becomes clearer.
 - Locations with very different climates can be compared.
- The objective is to estimate the large-scale average change—not to measure every location.

The Global Surface Is Warming

- Global average surface temperature increased by approximately **1.10°C** between:
- **1850–1900 and 2009–2018**
- Some years are warmer or cooler than the previous year.
- These short-term fluctuations occur around a rising long-term trend.
- Most of the warmest years occur in recent decades.
- **Climate change is identified from the long-term trend—not from one individual year.**



Source: Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.1. Gray shows annual anomalies, while blue shows the smoothed warming trend, relative to the 1951–1980 average.

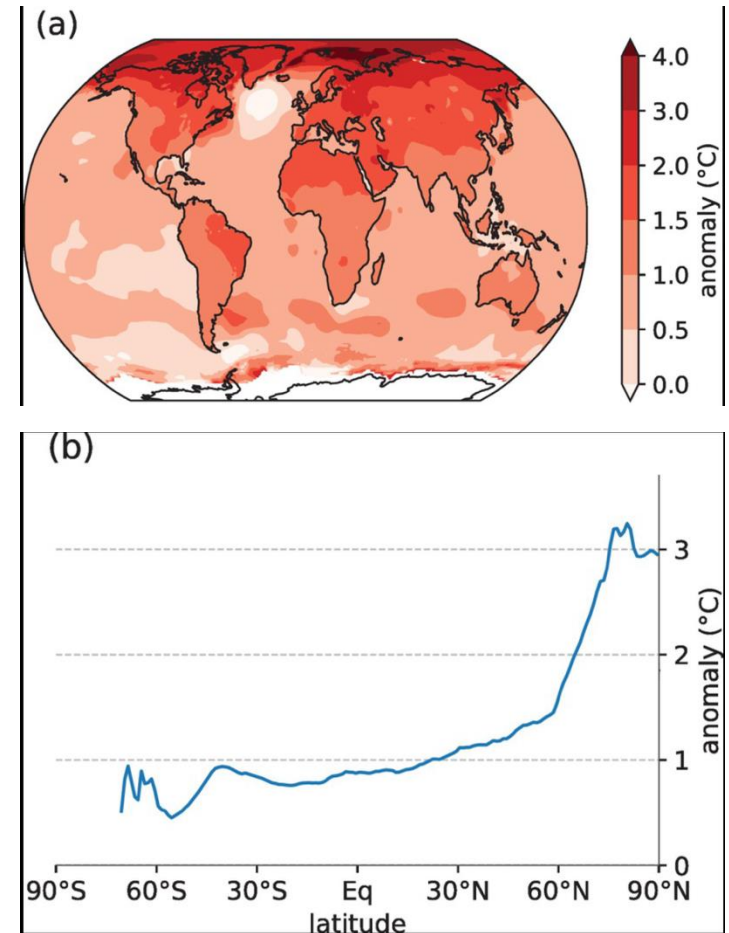
Warming Is Not Geographically Uniform

- Land generally warms faster than the ocean.
- The Northern Hemisphere has warmed more strongly, partly because it contains more land.

Q) Why is warming greatest in the Arctic?

- Arctic amplification
 - Melting snow and sea ice
 - darker ocean surfaces
 - more solar energy absorbed
 - additional warming and melting.

Result: The Arctic warms several times faster than the global average.



Did Global Warming Really Stop in 1997?

A 2012 article examined the claim that global temperatures showed “**no discernable rise**” between August 1997 and August 2012.

Discuss

- Does this mean global warming stopped?
- Why might 1997 have been selected as the starting year?
- Is 15 years sufficient to identify a climate trend?

SCIENCE Did Global Warming Really Stop in 1997?

By · LiveScience

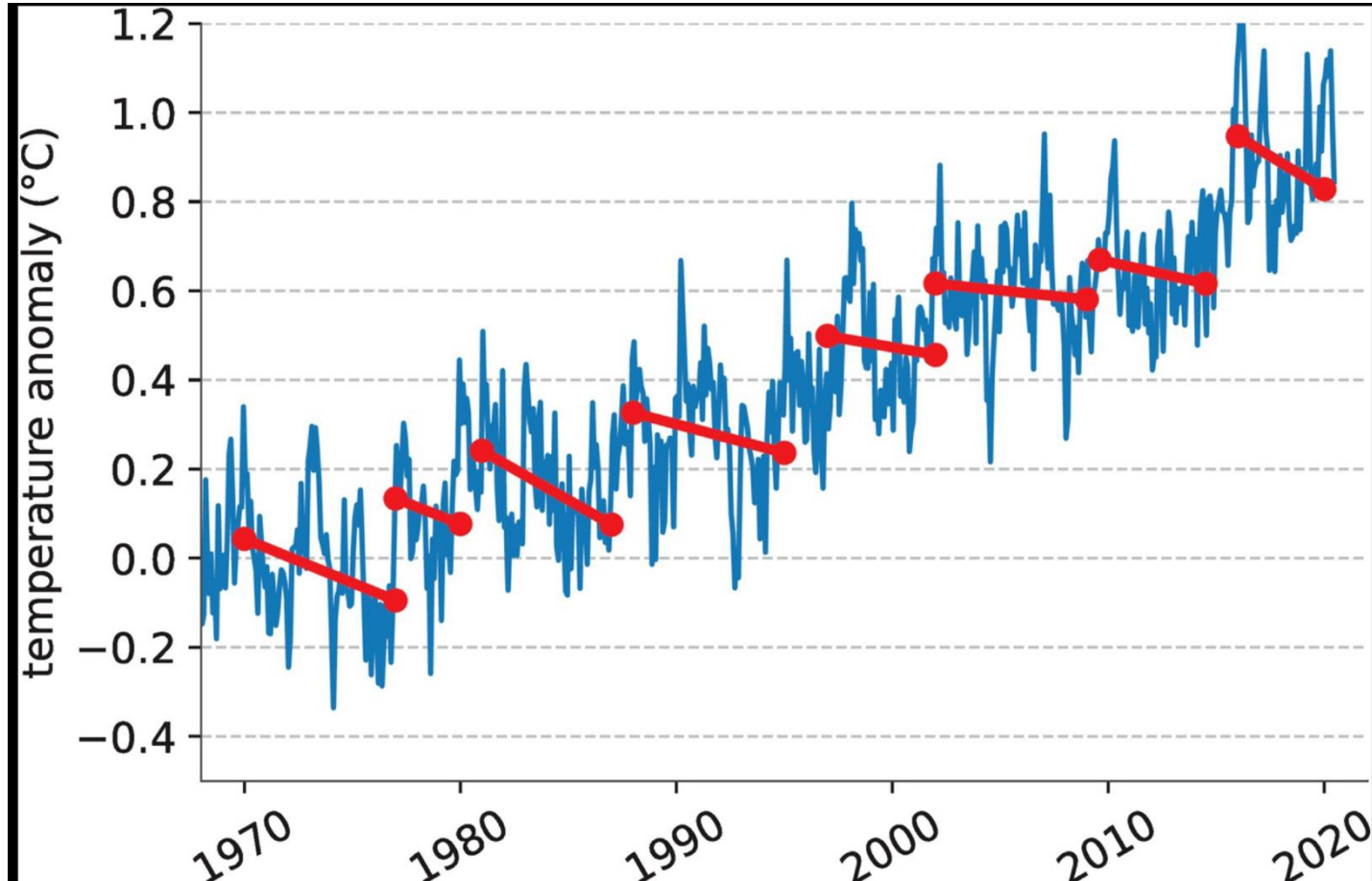
Published October 16, 2012 9:16am EDT | Updated October 21, 2015 4:56am EDT



Claims global warming stopped 15 years ago are based on "cherry-picked" data and don't account for natural fluctuations in climate, according to climate scientists responding to an article that appeared Saturday (Oct. 13) in the British newspaper, The Daily Mail.

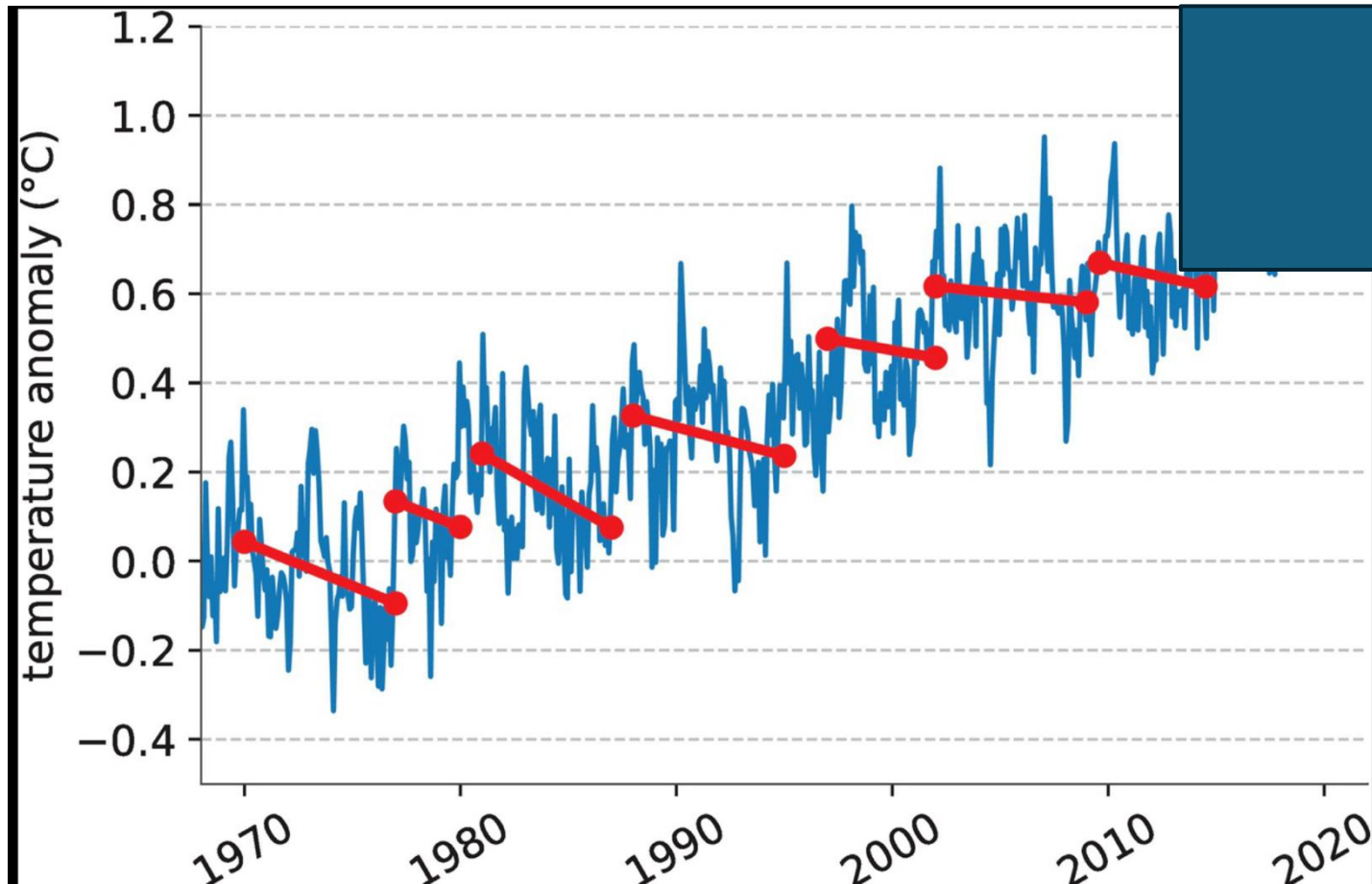
The article cites combined global land and sea-temperature data compiled by British climate researchers, claiming that between August 1997 and August 2012, ["there was no discernable rise](#) in aggregate global temperatures."

What Does the Record Show?



Source: Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.5. Fox new article published on 2015

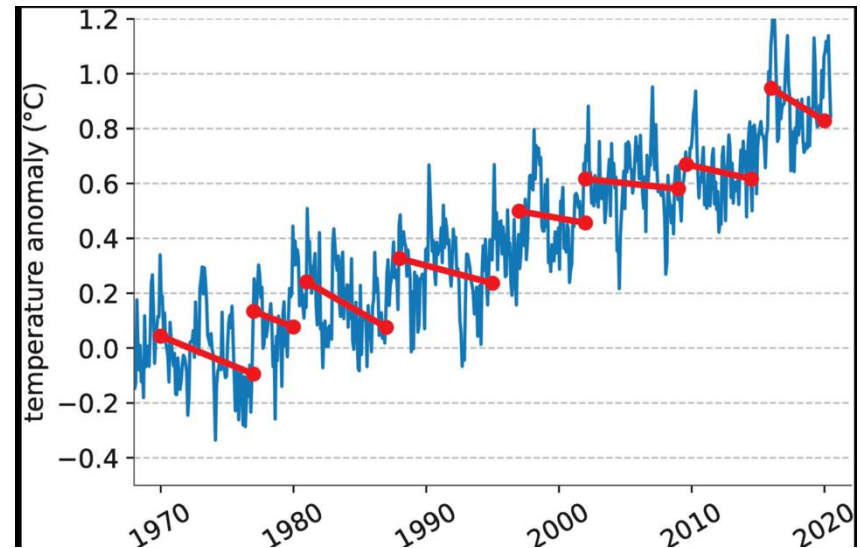
What Does the Record Show?



Imagine it's 2015, do you agree with Fox News ' claims?

How Can the Same Data Tell Different Stories?

- Starting with an unusually warm year and ending with a cooler year can create an apparent cooling trend.
- The reverse can exaggerate warming.
- Selecting dates that support a particular conclusion is called cherry-picking.
- Long-term climate trends should not be inferred from short, selected periods.



Dessler, A. E. (2021), Introduction to Modern Climate Change, 3rd ed., Chapter 2, Figure 2.5: Short segments can support misleading claims.

Why Does Temperature Fluctuate Around the Trend?

- The long-term warming trend includes temporary rises and falls:
 - El Niño often causes temporary global warming.
 - La Niña often causes temporary global cooling.
 - Volcanic eruptions can temporarily cool the climate by reflecting sunlight.
- A cooler year does not mean that the long-term warming trend has reversed..

Predict the Evidence of a Warming Climate

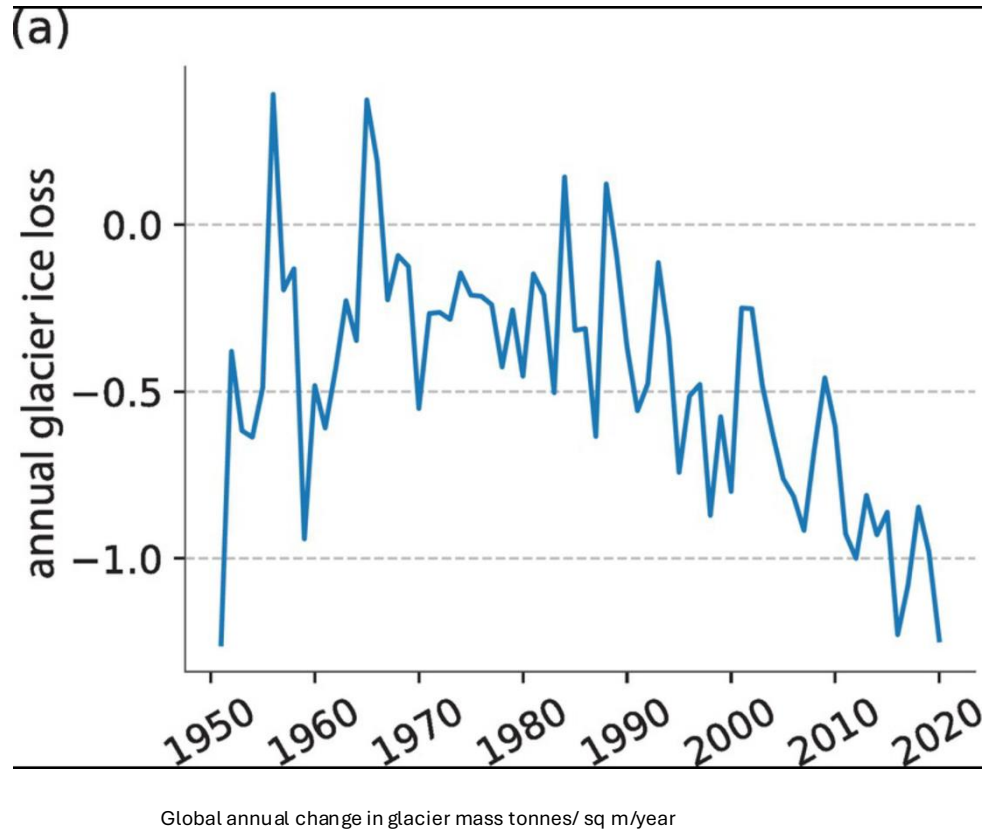
- Before examining the observations, predict how a warming climate should affect:
 1. Mountain glaciers
 2. Arctic sea ice
 3. Ocean heat content
 4. Global mean sea level
- For each indicator, write the expected direction of change:

Increase ↑

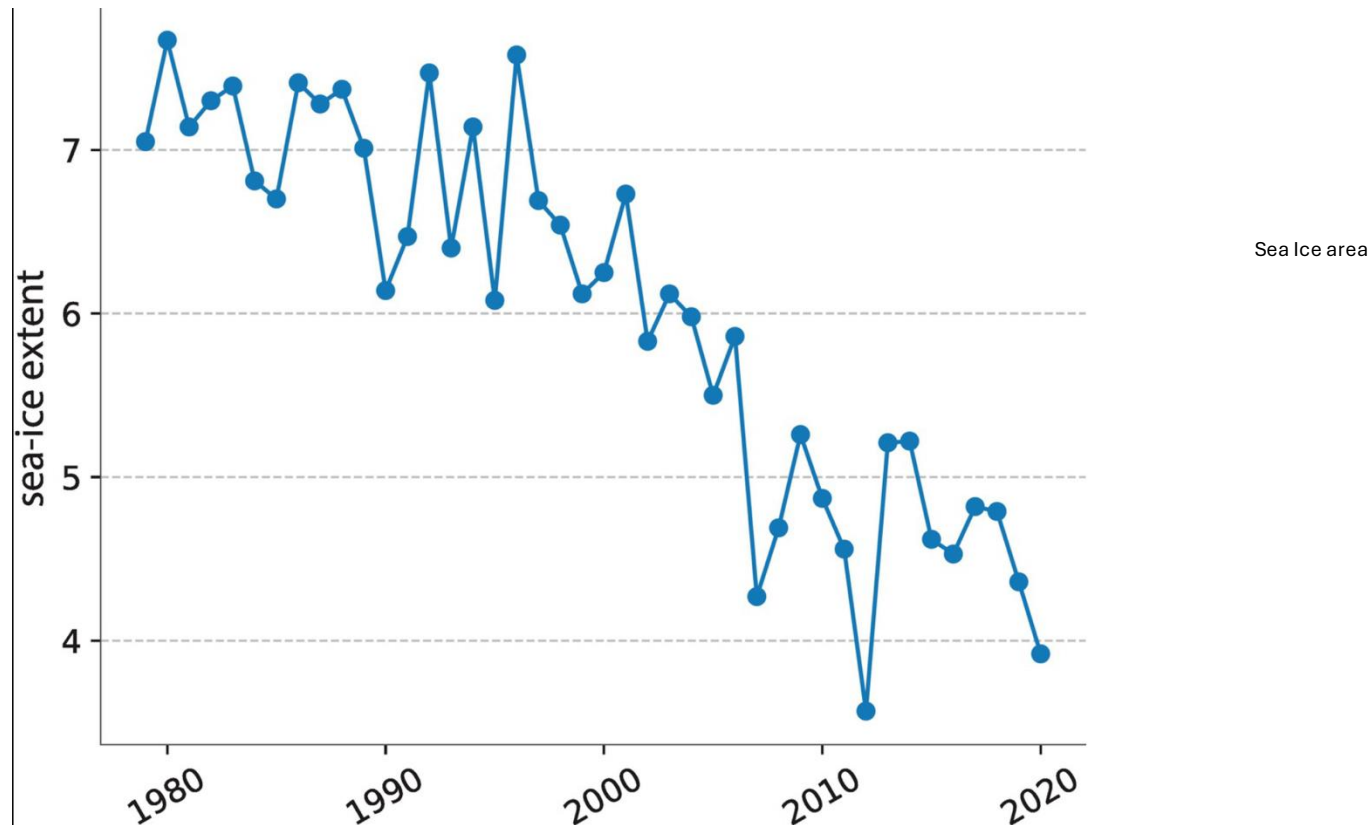
Decrease ↓

No clear change →

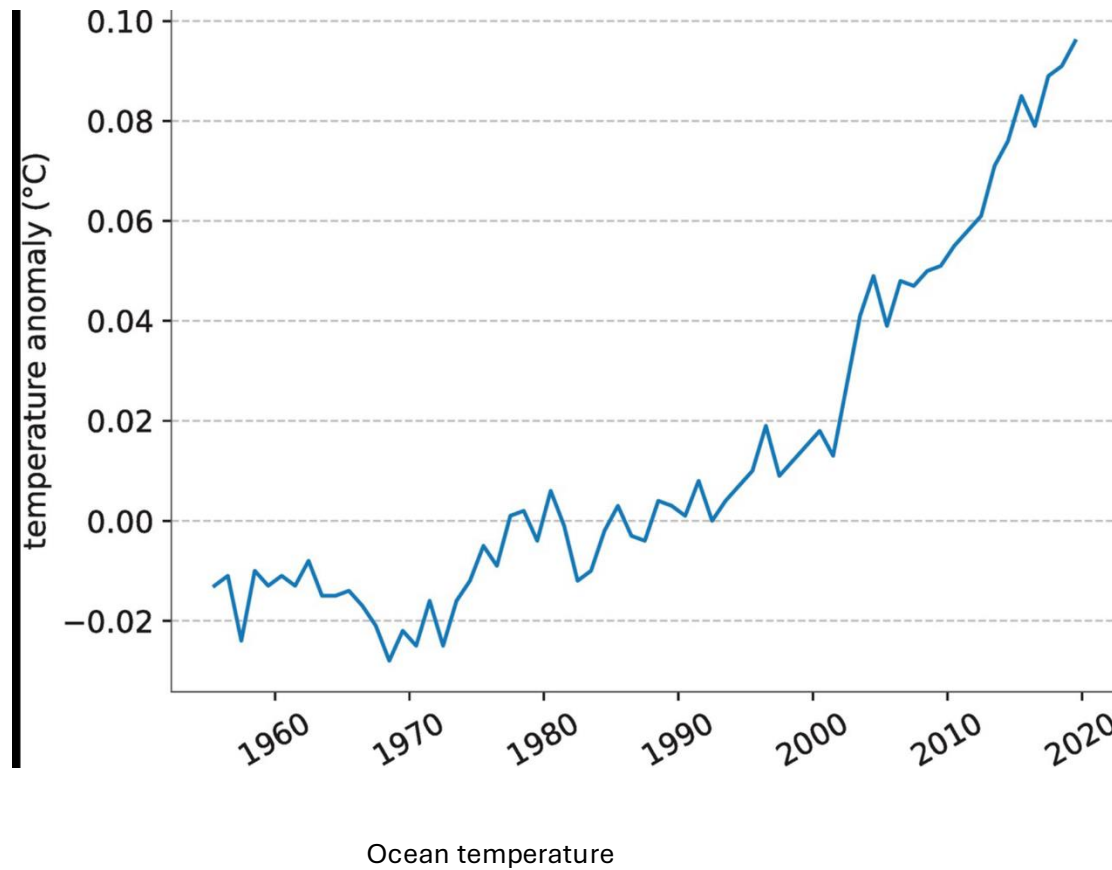
1. Mountain glaciers and major ice sheets are losing mass.



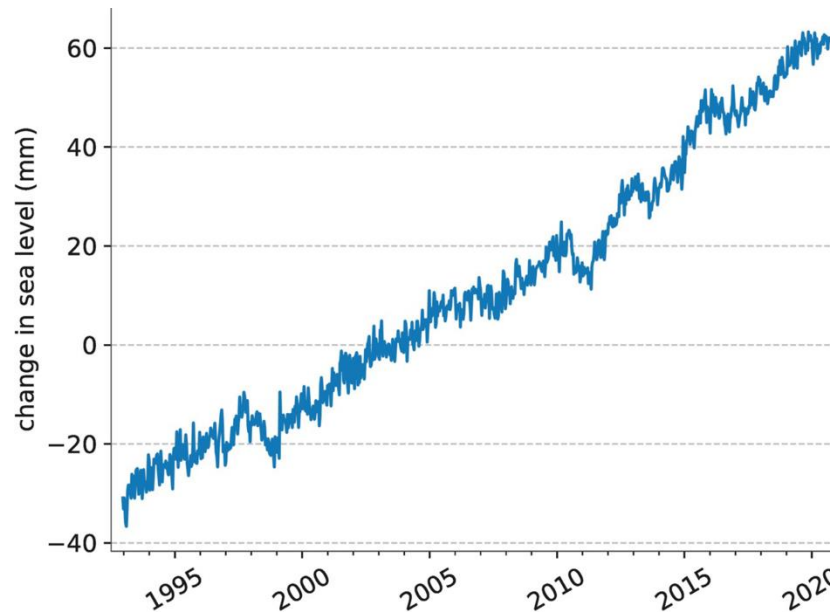
2. Arctic September sea-ice extent is declining.



3. The upper ocean is accumulating heat



4. Global mean sea level is rising.



These independent indicators tell the same story. An error in surface-temperature measurements cannot produce matching trends across all these systems.

How Do We Know the Climate Is Changing?

- A single heatwave, cold winter or regional event is not sufficient evidence. A short temperature record can also be misleading when its start and end dates are selectively chosen.
- Our confidence comes from:
 - Long-term changes in climate statistics
 - Observations covering many regions
 - Independent datasets and measurement systems
 - Consistent changes in temperature, glaciers, sea ice, ocean heat and sea level
- **Key takeaway:** Climate change is established from multiple independent lines of evidence—not from any single event, location or dataset.

Which line of evidence do you find most convincing, and why?